Approval gates and idempotency

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Approval gates and idempotency before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Planning and action selection

Without opening the old lesson, explain one key idea from Planning and action selection. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Logs traces and privacy

Without opening the old lesson, explain one key idea from Logs traces and privacy. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Integration tests and fixtures

Without opening the old lesson, explain one key idea from Integration tests and fixtures. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Approval gates bind side effects to an actor, exact version and destination. Idempotency prevents repeated requests with the same key from repeating a committed effect.

The mock submission requires approval for request version v1 and destination D. Store key -> committed result. Reusing the same key with the same payload returns that result; the same key with different payload is rejected.

Practise with fading help

> Complete the supported Approval gates and idempotency trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Approval gate**s and idempotency. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Approval gates and idempotency explanation against the supplied facts before discussing it.

Use the worked case for **Approval gate**s and idempotency. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

> Key J already committed room A. A retry uses key J but room B. What should happen?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Approval gates and idempotency facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 Trace the worked case

> Trace the worked case. Apply the supplied Approval gates and idempotency facts and keep your reasoning.

Approved send v1 to D uses key K, commits once, then caller retries K with the identical payload. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Approval gates and idempotency facts and keep your reasoning.

A report omits this qualification: A key prevents duplicate application effects only within the implemented persistence and downstream guarantees; it is not universal exactly-once delivery.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Approval gates and idempotency facts and keep your reasoning.

> Key J already committed room A. A retry uses key J but room B. What should happen?

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Approval gates and idempotency facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Approval gates and idempotency to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled **approval gate**s and idempotency **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **approval gate**s and idempotency.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Approval gates and idempotency

> One approval, one reservation. A desk tool needs approval for the exact request and deduplicates repeated request keys. Explain the labelled step before checking the changed condition.

One approval, one reservation. A desk tool needs approval for the exact request and deduplicates repeated request keys.

K9 reserves desk 4 and is approved once. Two copies of K9 arrive.

AI ACADEMY / AI-6/15

Approval gates and idempotency

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check approval

The approved identifier and desk match K9.



2. Execute first copy

No result is stored yet, so perform the approved reservation once.



3. Handle duplicate

Return the stored K9 result without another reservation.

CHANGE ONE THING

A new K10 asks for desk 5 without approval.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the two deliveries of approved K9. How many reservations should occur, and why can a new K10 for desk 5 not borrow K9's approval?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

One approval, one reservation. A desk tool needs approval for the exact request and deduplicates repeated request keys.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Return the stored K9 result without another reservation.

> 1. Execute first copy

2. Check approval

3. Handle duplicate

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Handle duplicate Return the stored K9 result without another reservation. Approval answers whether an action may proceed.

Idempotency prevents repeated handling of that action from creating duplicate effects.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A new K10 asks for desk 5 without approval. Which conclusion is supported by the supplied facts?

1. Stop K10 at the **approval gate**.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Stop K10 at the **approval gate**. K9′s approval does not apply to a different action.

Repair a missing building block

Review Planning and action selection (ai-6/14). Goal: Explain the **mechanism** and terminology of planning and action selection. Short practice: Choose a permitted plan when no approved tower-height card exists.

What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?. Return to this week and explain what you changed.

Review Logs traces and privacy (ai-6/12). Goal: Explain the **mechanism** and terminology of logs traces and privacy. Short practice: Identify the purpose of R17, C8 and 240 ms in the trace.

Which field is deliberately omitted, and why would logging the whole question violate this design?. Return to this week and explain what you changed.

Review Integration tests and fixtures (ai-6/9). Goal: Explain the **mechanism** and terminology of integration tests and fixtures. Short practice: Follow request N through the fixture and adapter.

Which source and item should appear in the reply, and what precisely fails when the adapter returns M?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

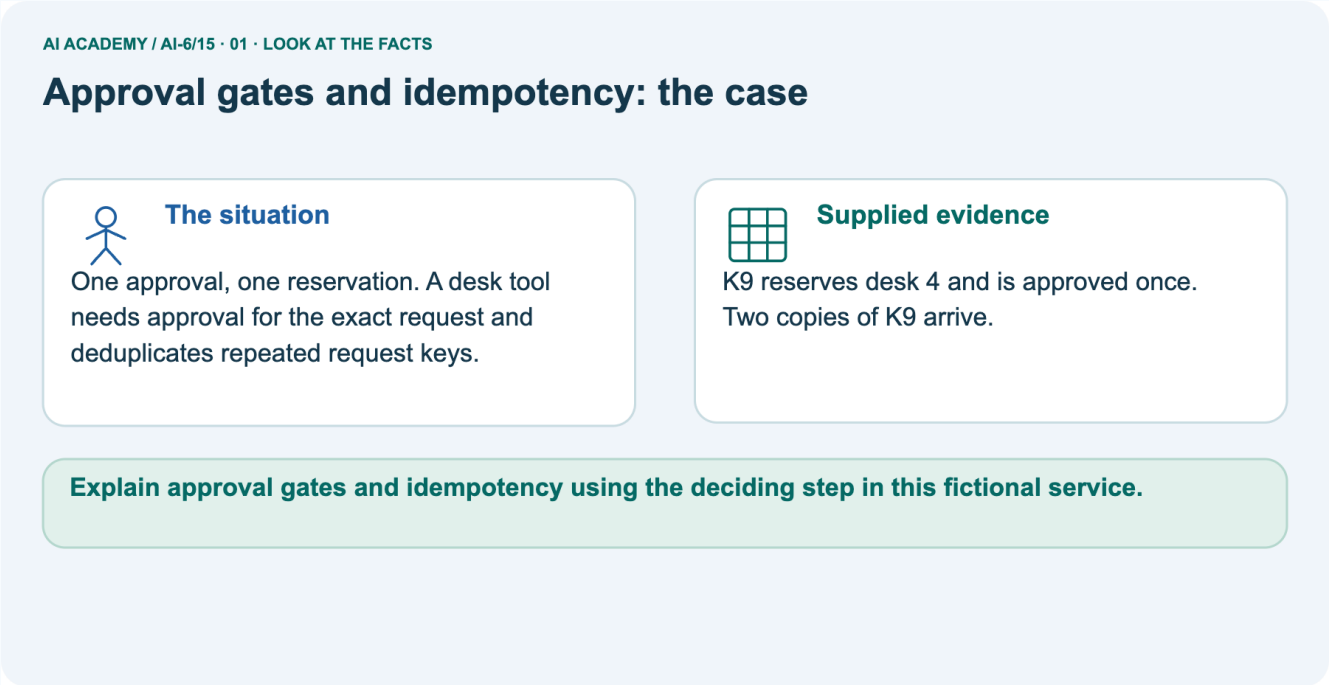
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Approval gates and idempotency picture from its labels, then record a relevant limit.

01. Approval gates and idempotency: the case



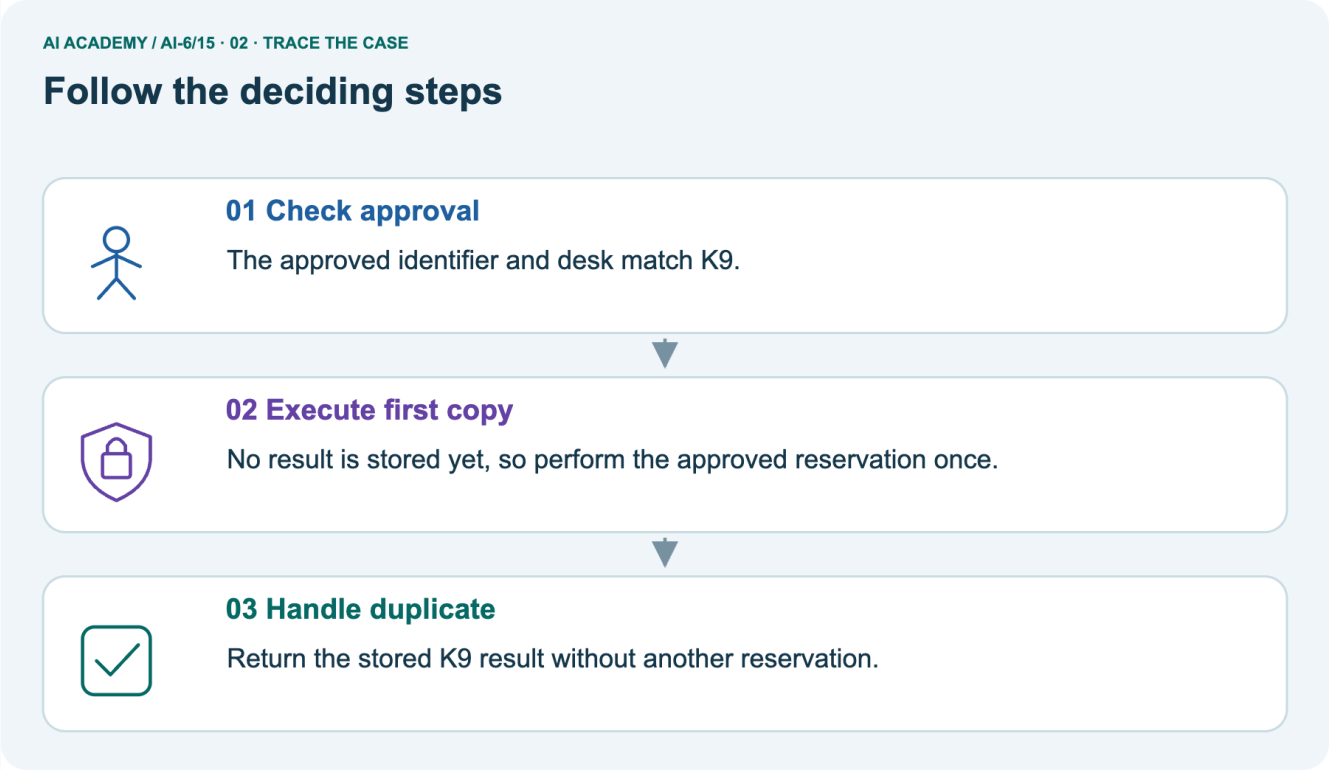
Follow the labelled picture with the matching explanation. Open the vector diagram

One approval, one reservation. A desk tool needs approval for the exact request and deduplicates repeated request keys. Supplied facts: K9 reserves desk 4 and is approved once. Two copies of K9 arrive.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



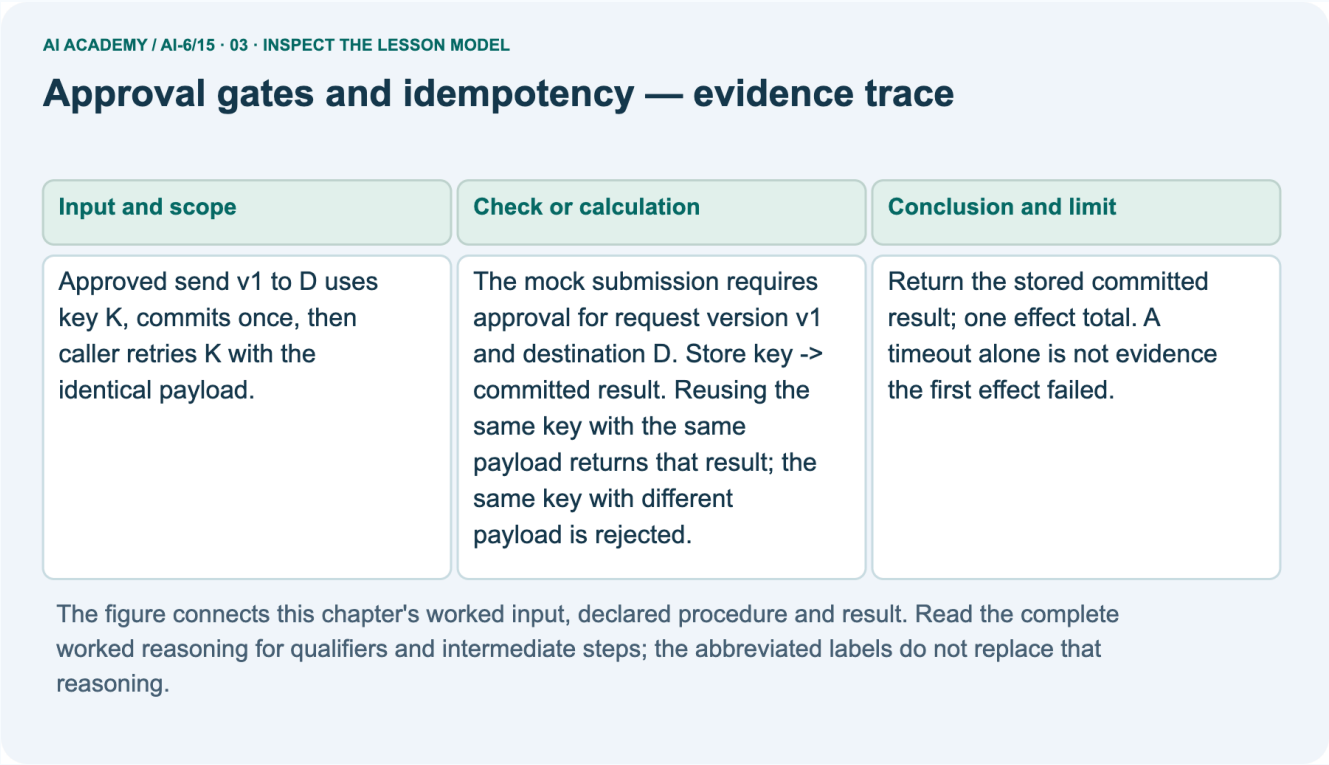
Follow the labelled picture with the matching explanation. Open the vector diagram

Check approval: The approved identifier and desk match K9. → Execute first copy: No result is stored yet, so perform the approved reservation once. → Handle duplicate: Return the stored K9 result without another reservation.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Approval gates and idempotency — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Approval gates and idempotency — **evidence** trace.

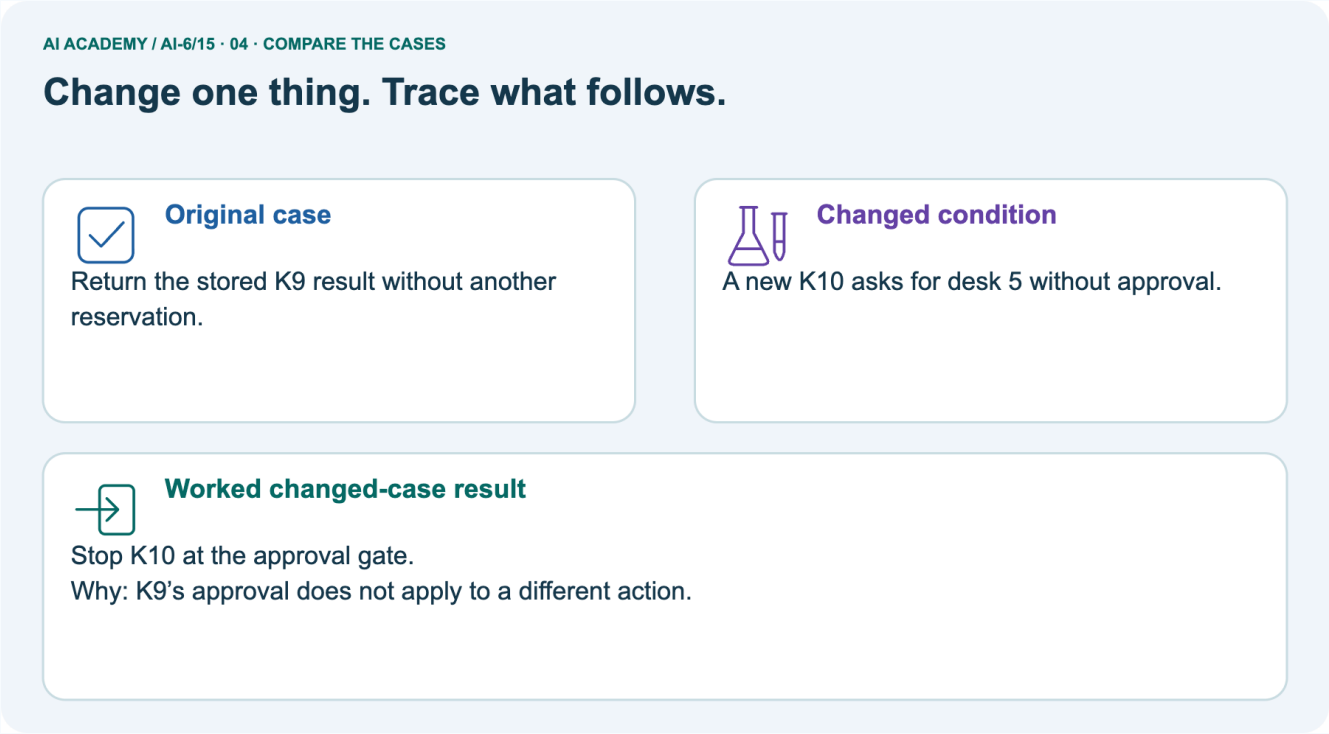
Approved send v1 to D uses key K, commits once, then caller retries K with the identical payload. | The mock submission requires approval for request version v1 and destination D. Store key -> committed result.

Reusing the same key with the same payload returns that result; the same key with different payload is rejected. | Return the stored committed result; one effect total. A timeout alone is not **evidence** the first effect failed.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

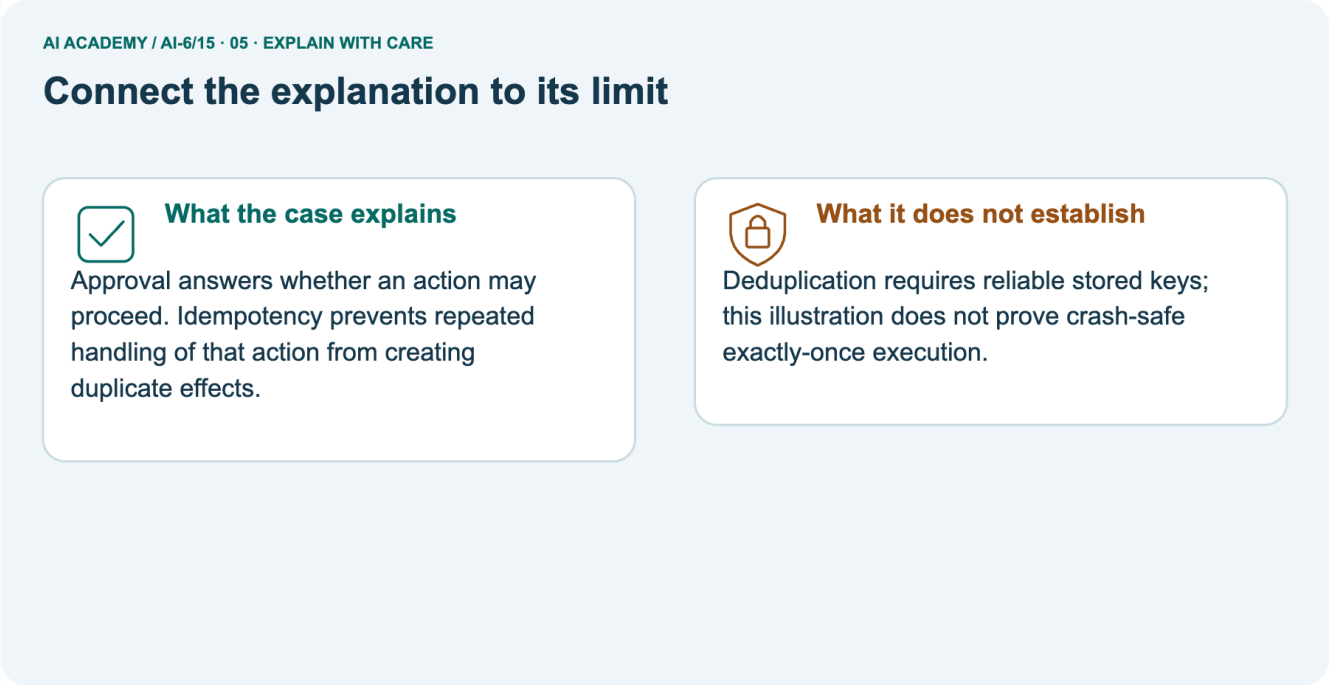
Original case: Return the stored K9 result without another reservation. Changed condition: A new K10 asks for desk 5 without approval. Expected result: Stop K10 at the **approval gate**.

Reason: K9's approval does not apply to a different action.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

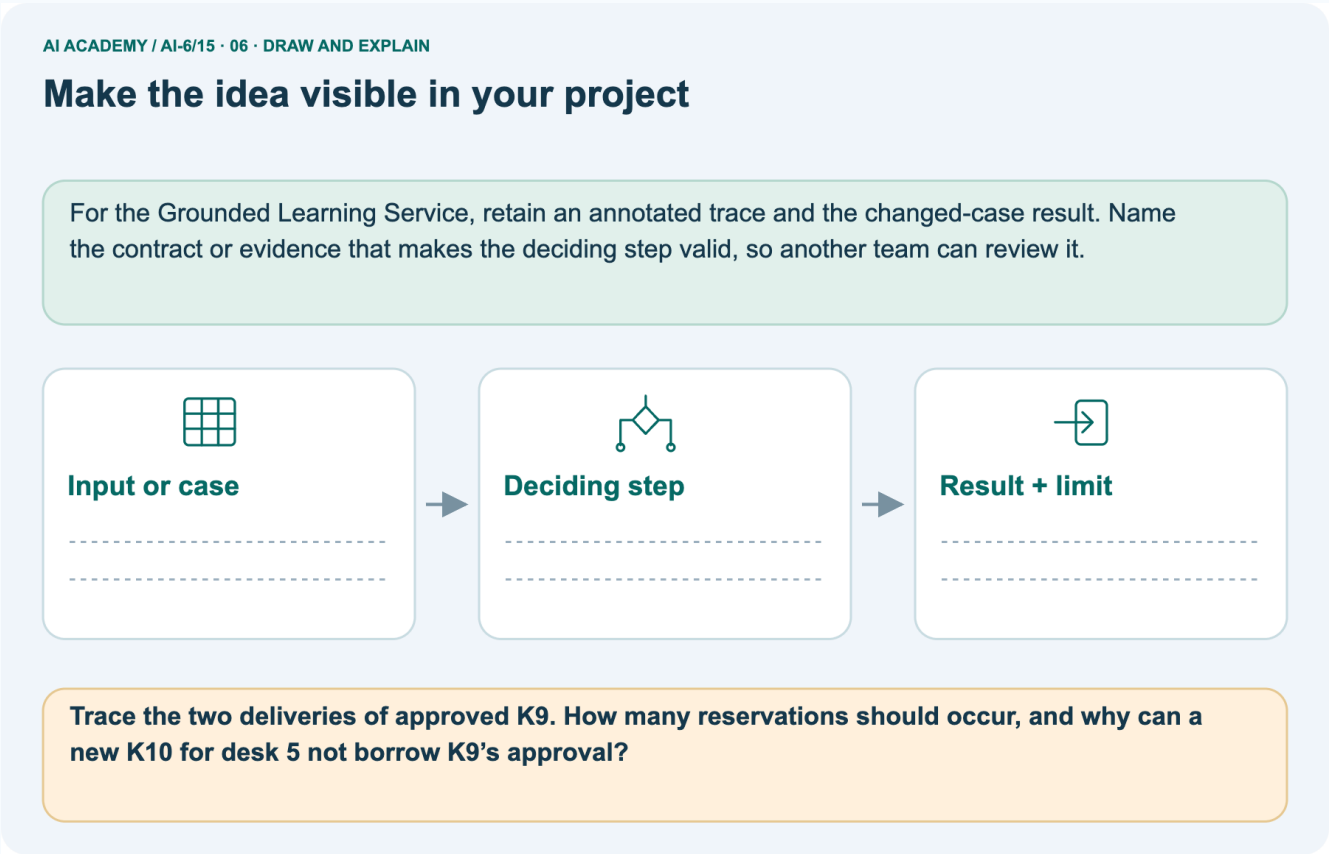
Approval answers whether an action may proceed. Idempotency prevents repeated handling of that action from creating duplicate effects.

Limit: Deduplication requires reliable stored keys; this illustration does not prove crash-safe exactly-once execution.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Trace the two deliveries of approved K9. How many reservations should occur, and why can a new K10 for desk 5 not borrow K9’s approval?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.